

The J-space in OLMo-3-32B-Think: an open-weights replication of “Verbalizable Representations Form a Global Workspace in Language Models”

special-lab-1, interpretability labs

July 26, 2026

run 2026-07-25_1726 · lens jlens @ 581d3986 · seeds 0

v2 delta run in progress (§9): results below marked [v2] update as instruments land

Abstract

We replicate the core claims of Anthropic’s July 2026 “global workspace” paper on **Olmo-3-32B-Think**, a fully open (weights + training data) reasoning model, using the reference Jacobian-lens implementation. The *geometry* replicates: a sparse, verbalizable subspace concentrated in the middle layers, whose directions are read by 73–94 downstream components (vs. 0 for random directions). Its *capacity* is roughly an order of magnitude thinner than reported for Claude ($\leq 0.67\%$ of activation variance vs. 6–10%; ~ 6 active concepts vs. ~ 25). The paper’s headline *causal dissociation does not replicate* at our doses: static J-space ablation is indistinguishable from random controls, while matched-dimension *non-verbalizable* high-variance directions cause the real selective damage. Exploiting the thinking variant, we add a result the paper could not test: the workspace does *not* contain the answer before reasoning begins (median rank 5613), but **leads the chain of thought by a median 46 tokens** during reasoning, and is loaded on demand when thinking is suppressed (rank 211 at the answer position, collapsing monotonically L30→L60). An eval-awareness probe finds a highly consistent “being-evaluated” readout direction ($p < 10^{-3}$ on all probed layers) that is semantically legible yet behaviorally inert under ablation.

1 Setup

Model. allenai/Olmo-3-32B-Think (64 layers, $d = 5120$), bf16 on one 96 GB RTX PRO 6000 Blackwell — full-precision tier: no quantization, no gradient checkpointing. **Lens.** Reference jlens package (anthropics/jacobian-lens), fit on 120 WikiText-103 prompts $\times$ 128 tokens (the neuronpedia reference recipe; jlens README reports saturation near 100; the paper used 1000). Source layers: dense middle band L20–44 (step 2) + early/late controls {4,8,12,16,48,52,56,60}; target L63; four 30-prompt slices merged via JacobianLens.merge. Fit cost 5.2 h (~ 157 s/prompt, peak 80.6 GB); per-prompt max $\|J\|/\sqrt{d}$ stable at 0.94–0.95 and running-mean deltas fell $\sim 1/n$ (converged). **Dictionary.** Concept i at layer ℓ read through the RMSNorm-gain linearization ($W_U \odot g$) J_ℓ , rows normalized. **Sanity gate.** Easy single-token factual probes: 17/21 hit rank ≤ 20 mid-band for *both* J-lens and logit lens (ceiling). The discriminative check, multihop bridge-entity readout at the final prompt token ($n = 60$): J-lens pass@1 **0.283** vs. logit-lens 0.200 (+41% rel.); pass@20 0.547 vs. 0.586 — J-lens sharpens the rank-1 readout without expanding the top-20 shortlist, the same “replicates, less clean” flavor as Nanda’s Qwen 3.6 27B replication.

2 Descriptive geometry: the workspace exists, ten times thinner

Sparse nonnegative gradient-pursuit decomposition of 200 diverse prompts (factual QA, chained arithmetic, code/SQL, prose) onto the dictionary:

- **Sparsity.** Median active concepts per (position, layer): **6** at $\theta=0.01$, 3 at $\theta=0.02$, $k_{90} \approx 4$, peaking at L36–42. Paper: ~ 10 –25.
- **Variance share.** $\leq 0.67\%$ of centered activation variance inside the band (0.91% at L60, unembedding-adjacent). Paper: 6–10%. The capacity gap ($\sim 10\times$) is threshold-robust.

- **Localization.** The cleanest signature is top-1 persistence across adjacent positions: 0.19–0.20 at L32–40 vs. 0.03–0.08 at both ends — an inverted U at 50–66% depth (paper band: 33–92%).
- **Broadcast — with a catch.** Top J -directions at L24/32/40 are read (input-projection alignment $z > 3$) by **73–94** downstream components vs. **0** for matched random directions ($p \approx 10^{-24}$). But matched-dimension top *non- J* principal components are read by 77–125: wide readout is a property of high-variance directions generally, not of verbalizable ones specifically.

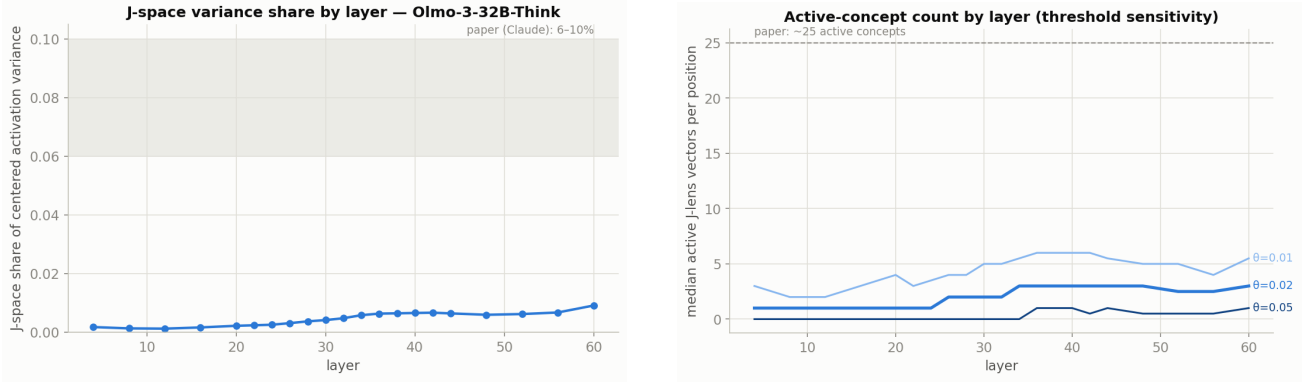


Figure 1: **Left:** J -space share of activation variance by layer; shaded band = paper’s Claude range (6–10%). **Right:** median active-concept count by layer under three activation thresholds; dashed line = paper’s ~ 25 .

3 Causal ablation: the dissociation does not replicate

Battery: 2-hop probe-swap facts ($n=60$), chained arithmetic ($n=30$), 3-table SQL with join-key checks ($n=30$), vs. one-hop facts, grammatical minimal pairs, held-out prose NLL. Conditions (band L20–44, doses $k \in \{10, 20, 40\}$ dims/layer, 2000-resample bootstrap CIs): static top- k J -span; matched random subspace; matched top non- J PCs; paper-faithful per-token dynamic top-10.

condition	2-hop	2-hop Δ lp	arith	SQL	1-hop	grammar	NLL
none (baseline)	0.58	—	0.87	0.67	0.60	1.00	2.71
J -space dyn top-10	0.00	−23.0	0.00	0.00	0.00	0.30	24.34
J -span $k=20$	0.52	−0.17	0.93	0.67	0.60	1.00	2.85
random $k=20$	0.52	−0.08	0.90	0.67	0.60	1.00	2.75
non- J PCA $k=20$	0.28	−0.86	1.00	0.00	0.70	1.00	3.45

Table 1: Accuracy (and 2-hop answer-logprob delta vs. baseline) under ablation; $k=10/40$ follow the same pattern. Static J -span $\approx$ random everywhere; the *non-verbalizable* control causes the selective damage.

Three observations. (1) The paper-faithful *dynamic* condition removes whatever the model is currently computing — output degenerates to asterisks, NLL 2.71→24.34 — reproducing the “live-computation confound” Nanda flagged; it cannot count as evidence for a workspace. (2) Static J -span ablation is statistically indistinguishable from random subspaces on every task *and* on the finer answer-logprob instrument (Δ −0.15... −0.18 vs. −0.01... −0.11; CIs straddle 0). (3) The non- J high-variance control produces the only structured damage — and generation samples show its signature is *register destruction*: SQL completions lose the formal register entirely (SELECT the user’s city and the user’s country, and the user’s name, ...) while arithmetic content survives. At these doses, OLMo’s causal load sits in generic high-variance directions, not in the verbalizable subspace.

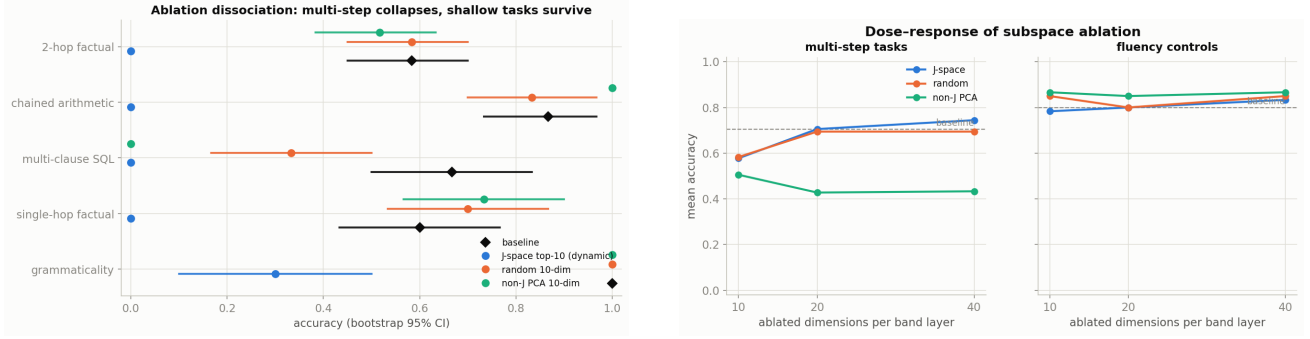


Figure 2: **Left:** per-task accuracy with 95% bootstrap CIs under the $k=10$ conditions. **Right:** dose-response for multi-step vs. fluency task groups.

4 The thinking-model angle: the workspace leads the chain of thought

Olmo-3-32B-Think emits explicit reasoning tokens, so the paper’s identity claim (workspace content = verbalizable reasoning substrate) becomes testable in time. 90 items (40 two-hop, 30 chained arithmetic, 20 SQL join-key); J-readout at L24/32/40 at every generation step; suppressed-CoT variant closes the template’s open `<think>` immediately.

- **Pre-CoT anticipation: null.** At the final prompt token the answer’s median J-rank is **5613** (10% of items ≤ 20); logit lens 11610 (0%). The workspace does not precompute the answer before reasoning.
- **Intra-CoT anticipation: strong.** The answer enters a read layer’s top-8 during thinking in 82–100% of items, and on traced items ($n=34$) the workspace holds the answer a **median 46 steps before the CoT first states it** (2-hop 48.5, SQL 38, arithmetic 9.5; workspace first in 91% of items).
- **Answer-time loading.** Forcing an immediate answer (closed `</think>`, zero reasoning tokens) pulls the answer’s median rank from 5613 to **211** (23% ≤ 20), collapsing monotonically L30→L60: 2048 → 1549 → 519 → 305 → 118 → 63 → 10 → 6, with J-lens ahead of logit lens through L28–40. Suppression costs accuracy: 2-hop 0.95 → 0.82, arithmetic 1.00 → 0.90, SQL 1.00 → 1.00.
- **Faithfulness probe.** 116 events where a stable workspace top-1 word never appears in the nearby CoT (59/90 items) — but only 4 involve the final answer. The recurring divergent words are *discourse-mode* adverbs: **famously** (stable top-1 for 7–10 steps while the CoT recalls “Springfield ... birthplace of basketball”), **respectively**, **etc** during enumerations. We find no “workspace holds X, CoT argues Y, answer is X” case; the workspace appears to track the discourse mode of the current reasoning span rather than concealed content.

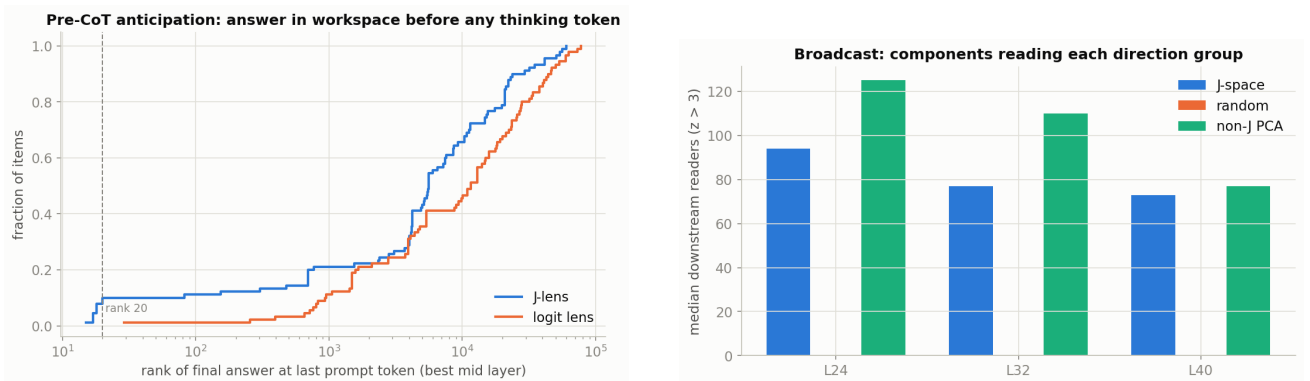


Figure 3: **Left:** ECDF of the answer’s rank at the last prompt token (best mid layer), J-lens vs. logit lens — anticipation is absent in both. **Right:** median downstream readers per direction group (broadcast).

5 Stretch: an eval-awareness direction, legible but inert

Twenty tasks rendered neutrally vs. prefixed with an explicit “formal evaluation, you are being tested and scored” framing; paired J-readout differencing at the final prompt token, seven band layers. A consistent direction emerges far beyond the pre-registered criterion: sign-flip permutation $p < 10^{-3}$ at *every* layer, mean pairwise cosine of paired deltas 0.48–0.71, mean-delta norm growing with depth (22→134). From L32 upward its top decoded tokens are literally the evaluation lexicon (**evaluation, assessed, assessing, eval**); at L24–28 the associates are obliquer (**monitored, internally, careful**). Yet the gated mini-ablation — projecting out the mean Δh direction during generation — changed essentially nothing. Because the framing states the cue words outright, this result is best read as a clean demonstration that the J-space *verbalizes context*, not as evidence of a latent “I am being tested” state; the confound-breaking variant (implied, unstated cues) is future work.

6 Side-by-side

	Claude (paper)	Qwen 3.6 27B (Nanda)	Olmo-3-32B-Think (ours)
Lens recipe	1000 prompts	25 Pile prompts	120 WikiText prompts, 21 layers
Active concepts	~10–25	not headline	6 ($\theta=0.01$); 3 ($\theta=0.02$)
Variance share	6–10%	—	$\leq 0.67\%$ ($\sim 10\times$ thinner)
Localization	33–92% depth	bands “less clean”	inverted U, peak 50–66% depth
Broadcast	wide readout	—	73–94 vs. 0 random; but non-J PCs 77–125
Causal dissociation	top-10 ablation kills multi-hop, spares shallow	flagged as possibly confounded	null : static J $\approx$ random; non-J PCs do the damage; dynamic = confounded collapse
Workspace vs. CoT	identity claim; CoT partially rescues	poetry/arith evals failed	pre-CoT null; leads CoT by 46 steps ; answer-time loading under suppression

Table 2: Replication scorecard. Paper/Nanda entries as pinned in the lab’s PLAN/LOG from the 2026-07-25 fetches.

7 Limitations

(1) Dictionary uses the RMSNorm-gain linearization ($W_U \odot g$) J_ℓ ; the final block’s full nonlinearity is not inverted. (2) Fitted band 20–44 (33–70% depth) $\subset$ the paper’s 33–92%; the suppressed-CoT profile (rank still collapsing L44→60) suggests unfit late-band structure. (3) Active-concept thresholds are not perfectly commensurable with the paper’s pursuit coefficients; the variance-share gap is the firmer capacity statement. (4) The non-J control is top-variance, not variance-*matched*; its larger effect partly reflects more energy removed — the single most important follow-up. (5) The dynamic ablation deletes live computation (Nanda’s confound); a frozen prompt-selected top- k is the clean middle instrument, not yet run. (6) WikiText corpus (not Dolma); 120 fit prompts; single seed; modest n (30–60/task); CoT tasks near ceiling compress suppression contrasts. (7) Phase-5 framing states its cue words (semantic echo); its ablation was inert.

8 What to run next

(1) Variance-matched non-J controls for broadcast and ablation — energy vs. content. (2) Frozen prompt-selected top-10 ablation during generation. (3) CoT-rescue: static ablations under think-mode prompting (the paper predicts externalization rescues; our suppression asymmetry predicts it here). (4) Fit L46–62 and redo the readouts where answer formation actually happens. (5) Dolma-corpus lens — the scientific-legibility payoff of doing this on OLMO. (6) **famously**-attractor forensics across models. (7) Implied-cue eval-awareness with a behavioral endpoint.

9 The v2 delta run: resolving the open instruments

The v1 verdict left a live question: *is the causal null a property of OLMo or of our instruments?* The published Qwen comparison is narrower than "Qwen has a J-space" — its readout evals replicated while its causal interventions were flagged by the replicator himself as possibly confounded — and v1 has three known instrument gaps, each capable of producing a false null: the non-J control was not energy-matched (its damage could be an energy artifact), the static J-span is position-agnostic (could miss position-specific workspace structure), and the fitted band stops at 70% depth while v1’s own suppressed-CoT profile shows answer formation continuing to L60. The v2 run closes these in order (variance matching, frozen prompt-selected ablation, late-band fit), calibrates the cot-lead headline against a false-positive floor, then tests CoT rescue. Decision tree: if the clean instruments flip the causal result, v1’s null was harness; if they confirm it, the same instruments go to Qwen 3.6 27B (Neuronpedia’s published lens, our harness) to separate model from method.

9.1 [v2-P4] The cot-lead headline survives its false-positive floor

The v1 claim — workspace holds the answer a median 46 steps before the CoT states it — used a permissive detector (any case/space variant first-token in any of 3 read layers’ top-8, across hundreds of steps), so any word gets many chances to fire. Running the *identical* detector on matched foils (fig. 4): frequency-matched random content words fire at **0.06** (vs. **0.92** for answers) — the pure noise floor is 15× below the signal. Same-family wrong answers (other bridge entities, other join columns, nearby numbers) fire at 0.32 — but where they also appear in the CoT text their median lead is **3 steps** (vs. 46): family foils show up in the workspace *concurrently* with the text discussing them, exactly what a faithful context-verbalizing readout should do while the model enumerates candidates. Within items, the answer is detected earlier than *every* one of its foils in 66% of items (median answer detection rank 1). The 46-step lead is specific to the answer, not an artifact of detector permissiveness.

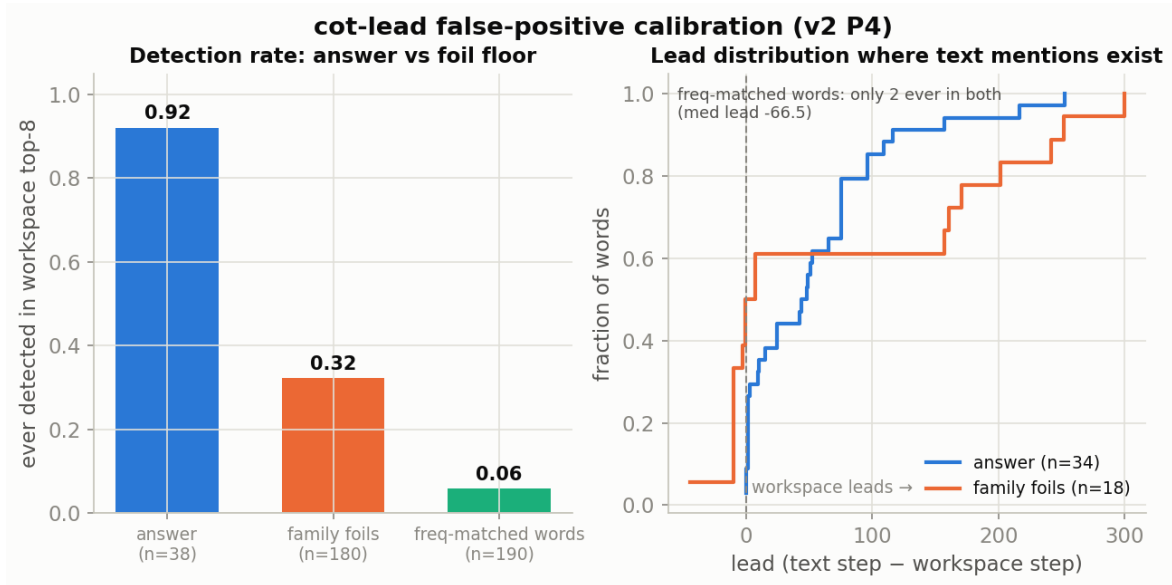


Figure 4: Foil calibration of the cot-lead detector. **Left:** detection rates — answers 0.92, family foils 0.32, frequency-matched words 0.06. **Right:** lead distributions where text mentions exist; the long positive lead is answer-specific.

9.2 [v2-P1a] v1’s controls were energy-mismatched by an order of magnitude — measured, then fixed

Streaming the raw residual stream over the 200-prompt descriptive set and measuring exactly what each ablation hook removes: the k-dim J-span carries 0.5–1.8% of raw activation energy (k=10/20/40), which equals **25–33 / 44–54 / 80–94 random dimensions** — or just **1–2 / 2–5 / 3–8 deep non-J PCs**. Both of v1’s rank-matched comparisons were therefore energy-confounded, in opposite directions: **random_k** removed $\sim 3\times$ less energy than

the J-span (which makes v1’s “J $\approx$ random” *conservative* — J removed more energy and still did no more damage), while `nonJ_pca_k` (top-k PCs) removed far *more* (its damage is suspect). The v2 matched controls hit the J-span’s removed energy to within 1% at every band layer and dose (fig. 5); the top PCs are so energy-fat that a contiguous *window* of deeper PCs (e.g. PCs 55–60) was needed at all 13 layers. The matched causal grid is running; its result decides whether v1’s “non-J does the real damage” was content or energy.

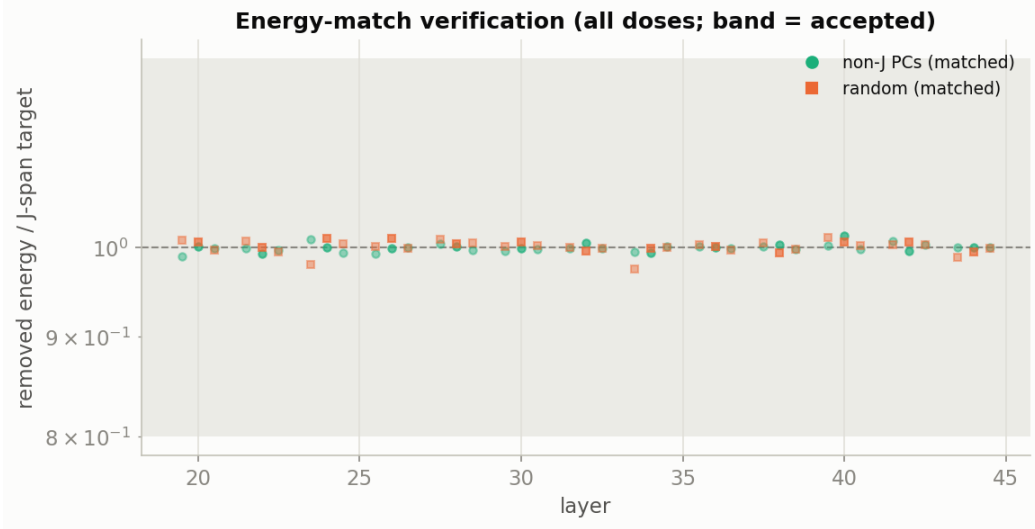


Figure 5: Energy-match verification: removed energy of each matched control relative to its J-span target (log scale), all band layers and doses. Every point sits on the 1.0 line, inside the pre-registered acceptance band.

9.3 [v2-P1b] At matched energy the causal null is clean: damage tracks energy, not content

The energy-matched grid (fig. 6) settles the question P1a raised. Every piece of v1’s apparent non-J selectivity disappears: SQL returns to baseline at every dose (0.667 vs. v1’s unmatched 0.000 — the “register destruction” was pure energy), the two-hop answer-logprob deltas shrink from $-0.45/ -0.86/ -0.93$ (CIs excluding zero) to $-0.01/ -0.17/ -0.15$ (every CI straddling zero), and prose NLL sits at 2.74–2.79 against a 2.71 baseline (v1 unmatched: up to 4.02). With the confound removed in both directions, the three groups are statistically indistinguishable from each other *and from baseline* at every dose up to 40 dims/layer:

2-hop $\Delta\log\text{prob}$ vs. baseline	k=10	k=20	k=40
J-span	−0.15	−0.17	−0.18
random (matched)	−0.11	−0.09	−0.08
non-J PCs (matched)	−0.01	−0.17	−0.15

Two consequences. First, the v1 causal verdict is *confirmed and strengthened*: on OLMo, statically removing the verbalizable subspace does nothing that removing an energy-equivalent arbitrary subspace doesn’t do — there is no content-specific causal load at these doses, and the earlier appearance of one was an instrument artifact we have now measured and eliminated. Second, v1’s own “J $\approx$ random” comparison was *conservative*: the J-span carries 2–5 $\times$ more energy than rank-matched random dims, so v1 was comparing a heavier J-removal against a lighter control and still finding parity. The last static escape hatch for the paper’s dissociation on OLMo is position-specific structure — frozen prompt-selected ablation (P2, running) and the late band (P3).

9.4 [v2-P2] The frozen instrument finds the first content-specific causal handle — and it is recall, not reasoning

The confound-free variant of the paper’s intervention: per item, one clean forward pass ranks the J-dictionary by activity over the prompt, the top-10 per band layer are *frozen*, and generation proceeds with that static per-item

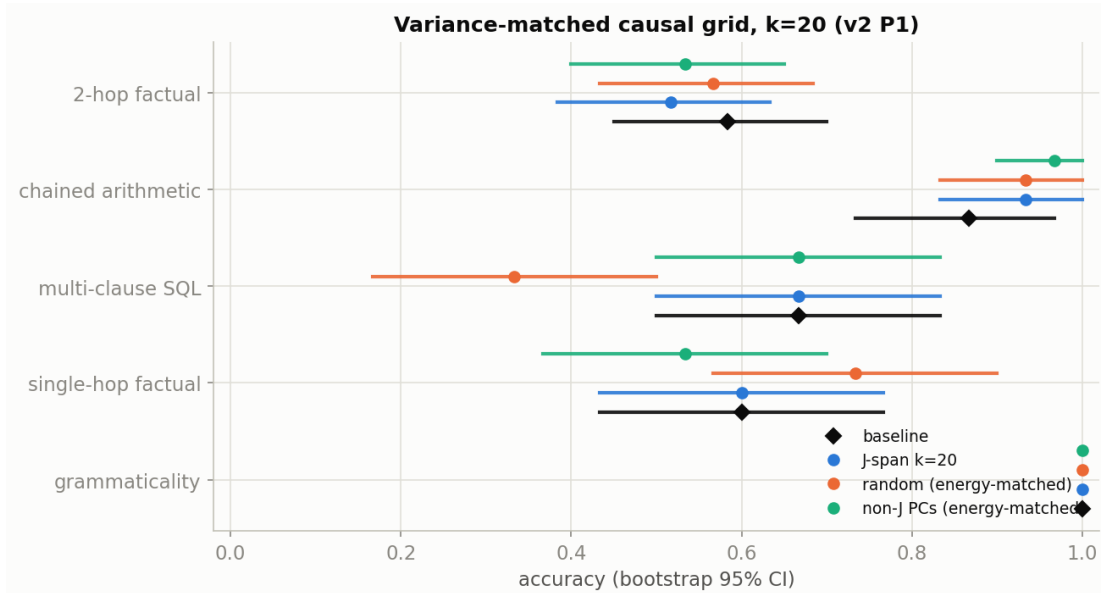


Figure 6: Energy-matched causal grid at k=20 (all doses in `v2_ablation_v2.json`). All three subspace families sit on top of each other and on baseline; contrast the v1 grid in §3, where the unmatched non-J control cratered SQL and two-hop.

projector. Two controls: the same mechanism on a random dictionary (`frozen_rand10`), and per-token live selection from a random dictionary (`live_rand10`, the matched-live control v1’s dynamic condition never had).

	2-hop	2-hop lp	1-hop	arith	SQL	NLL	grammar
baseline	0.58	−1.74	0.60	0.87	0.67	2.71	1.00
frozen J-10	0.23	−4.63	0.23	0.87	0.67	3.00	0.85
frozen rand-10	0.53	−1.78	0.53	0.90	0.33	2.79	0.95
live J-10 (v1)	0.00	−24.8	0.00	0.00	0.00	24.3	0.30
live rand-10	0.60	−1.82	0.67	0.87	0.33	2.80	0.95

Three readings, in order of importance. **(1) A control-clean causal effect, at last.** Freezing out the item’s top-10 J-directions deletes the retrieved fact: answer log-probability drops 2.9 nats (CI $[-5.49, -3.84]$ vs. baseline), recall accuracy falls $0.58 \rightarrow 0.23$ — while the identical mechanism on a random dictionary moves *nothing* (every cell within CI of baseline). Generation tasks are untouched and the stored verbatim samples under frozen-J are coherent, correct arithmetic. After P1 eliminated every subspace-level effect, the per-item selection is what matters: the J-space’s causal content lives in *item-specific* directions, not in any fixed span. **(2) It is not the paper’s dissociation.** Single-hop recall collapses *identically* to two-hop (both 0.23) — the paper’s signature required shallow tasks to survive. What the frozen instrument found is a *factual-recall content channel*, not a multi-step scratchpad: removing the item’s J-directions removes what the model knows, not how it chains. **(3) v1’s live-J catastrophe was J-specific after all.** Live top-10 selection from a random dictionary does nothing (0.60 two-hop, NLL 2.80), so the v1 lobotomy was not “removing whatever aligns with the current state” generically — though the dictionaries’ pool sizes differ (full vocab vs. 5120 rows), so alignment depth is a residual caveat.

Synthesis with P1: *static span removal does nothing at any matched dose; per-item frozen selection deletes facts; per-token live selection deletes the computation.* On OLMo the causal story of the J-space is content-carriage — causally real, control-clean, and increasingly precisely localized — but the workspace-as-reasoning-substrate claim remains unsupported. (The recurring SQL 0.33 cells in both random controls are the known 3-schema flaky cell; $n=30$ over 3 patterns.)

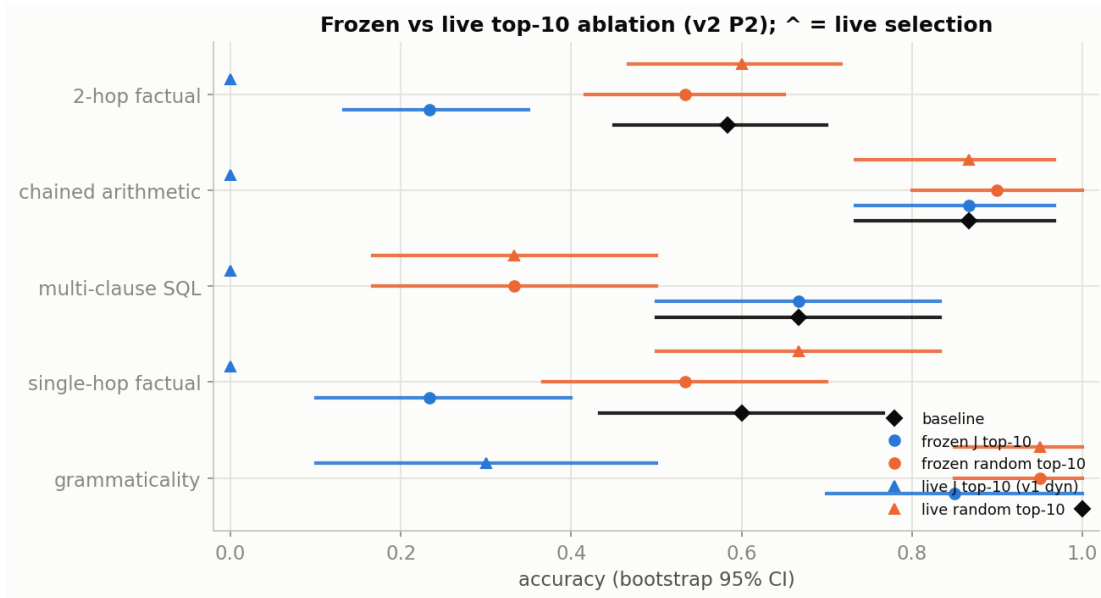


Figure 7: Frozen vs. live top-10 ablation (triangles = live selection). Frozen-J selectively destroys factual recall (1-hop and 2-hop alike) while its random-dictionary twin sits on baseline; live-J remains the uninterpretable collapse.

Artifacts

Everything regenerates from saved metrics without model access: run `dir interpret/special-lab-1/2026-07-25_1726/` (Drive) — `report/REPORT.md` (full prose), `report/summary.json` (headline numbers), `metrics/*.json` (per-experiment), `code/special_lab1/` (pipeline s0-s10 + PLAN/LOG). Model: `allenai/Olmo-3-32B-Think`. Lens engine: `github.com/anthropics/jacobian-lens @ 581d3986`.

References

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